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Patient-Reported Outcomes

Psychometric Performance of EQ-HWB and EQ-HWB-9 Self-Complete and Interviewer-Administered Versions in Literate, Low-Literacy, and Patient Populations in Indonesia



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ABSTRACT

Objectives: This study aimed to evaluate the psychometric performance, including test-retest reliability, ceiling/floor effects, and construct validity of the interviewer-administered (IA) and self-completion (SC) versions of EQ Health and Well-being instrument (EQ-HWB) and EQ-HWB-9.

Methods: A longitudinal survey in Bandung, Indonesia, recruited 300 respondents using stratified quota sampling, comprising 200 literate individuals, 50 with low literacy or illiteracy, and 50 patients. Respondents completed EQ-HWB, EQ-5D-5L, and Warwick-Edinburgh Mental Well-being Scale using either IA or SC modes. Psychometric analyses included ceiling/floor effects, convergent and known-group validity, and test-retest reliability (Gwet's AC2, intraclass correlation coefficients).

Results: The final sample (mean age 39.22 ± 15.81 , 62.67% female) demonstrated ceiling effects on EQ-HWB items ranging from 24.0% ("Exhaustion") to 86.33% ("Hearing"), with 5 items exceeding the 70% threshold. In contrast, EQ-HWB-9 showed lower ceiling effects, with only 1 item ("Getting around inside and outside") reaching the ceiling criterion (71.67%). Strong IA-SC agreement was observed (eg, AC2 = 0.95 for "Getting around inside and outside"). In the illiterate subgroup, item-level agreement remained strong, and instrument-level reliability was acceptable, supporting the feasibility of interviewer administration in low-literacy populations. Convergent validity was established with EQ-5D-5L and Warwick-Edinburgh Mental Well-being Scale. Known-group comparisons based on EQ-VAS (<80 vs ≥ 80) and patient status showed significant discrimination.

Conclusions: To our knowledge, this is the first study to validate the experimental IA versions of EQ-HWB and EQ-HWB-9 in Indonesia. The findings support their robust psychometric properties and feasibility for use in low-literacy and patient populations.

Keywords: EQ-HWB, EQ-HWB-9, illiterate, Indonesia, interviewer administered, literate.

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Introduction

The EuroQol Group developed an instrument to measure health-, social care-, and carer-related quality of life, called the EQ Health and Well-being instrument (EQ-HWB).^{1,2} Two experimental versions of the instrument are available: the long-form EQ-HWB, comprising 25 items and used as a profile measure, and the short-form EQ-HWB-9, a 9-item self-classifier derived from a subset of EQ-HWB items. Both are designed to capture a broad range of symptoms and functional aspects relevant to the general public, patients, families, social care users, and carers. The EQ-HWB and EQ-HWB-9 were developed through a rigorous multicountry process involving item generation and testing in the United Kingdom, United States, Germany, Argentina (Spanish

version), and Australia.^{3,4} Subsequent studies have reported on the measurement performance of the stand-alone instruments (EQ-HWB and EQ-HWB-9) in high-literacy countries, such as China, Italy, and Australia, particularly among general populations, patients, caregivers, and social care users.⁵⁻⁷ These studies have examined their distributional properties, construct validity, and, in some cases, test-retest reliability and responsiveness. However, to date, no validation studies have been conducted among illiterate or low-literacy populations,^{8,9} and an Interviewer-Administered (IA) version of EQ-HWB has not been developed or tested.

This study aimed to address the identified gap by conducting an investigation in Indonesia, where many population groups would benefit from an IA version of EQ-HWB because of high

illiteracy rates and the challenges of conducting face-to-face interviews. Although Indonesia reported a high literacy rate of approximately 96% for individuals aged 15 and older,¹⁰ 55% of those who completed school remained functionally illiterate, lacking the relevant skills for the labor market.¹¹ Consequently, the psychometric performance of EQ-HWB in both patients and general public populations, specifically regarding literacy levels and the use of an IA format, remains largely unexplored.

This study aimed to validate the Indonesian version of EQ-HWB and EQ-HWB-9 in literate, illiterate/low literacy, and patient groups. Notably, although EQ-HWB-9 responses are derived from a subset of EQ-HWB items, this study analyzed both instruments' psychometric performance as distinct constructs. It also aimed to provide initial evidence on the agreement between responses to the IA and SC versions of EQ-HWB in the literate group, assess test-retest reliability after 2 weeks across all groups, and compared the measurement properties of EQ-HWB/EQ-HWB-9 with well-established instruments, specifically the generic EQ-5D-5L and the Warwick-Edinburgh Mental Well-being Scale (WEMWBS). The results were expected to contribute to the growing body of psychometric evidence informing the development and application of these instruments.

Methods

Study Design and Respondents

Ethical approval was obtained from the Research Ethics Committee of Hasan Sadikin General Hospital (LB.02.01/X.6.5/284/2023), followed by the collection of longitudinal data. A minimum sample size of 239 patients was calculated to achieve 80% power in detecting a correlation coefficient greater than 0.18 between the corresponding domains of EQ-HWB and EQ-5D-5L instruments, using a 2-tailed significance level of 0.05. A target sample size of 300 was set to slightly exceed this minimum and allow for subgroup analyses.

This study used 3 respondents groups, (1) literate sample ($n = 200$), comprising individuals from the general population who had completed at least junior high school and were not recruited based on any specific health condition, (2) illiterate or low literacy sample ($n = 50$), comprising individuals from the general population who were either illiterate or had only completed or not finished primary education; this group was also not selected based on health status, and (3) patients ($n = 50$), comprising outpatients diagnosed with either tuberculosis or type 2 diabetes mellitus from a hospital in Bandung. Although only the patient group had confirmed diagnoses, all respondents completed the General State of Health (GH) questionnaire during the second follow-up session (2 weeks later for literate and illiterate groups, 1 month later for patients). To ensure diverse demographic representation, respondents were selected to include both genders and various age groups (below 30, 30-50, and above 50 years), particularly within the literate group.

A subsample from each group involved in the IA version of the EQ-5D-5L validation study was invited to participate in this EQ-HWB study. The mode of administration varied by instrument and respondent group. The EQ-HWB and EQ-HWB-9 were administered in both IA and SC formats depending on the group and study session. In the literate group, respondents alternated between SC and IA administrations of the EQ-HWB across 3 sessions to evaluate format effects. The EQ-5D-5L and WEMWBS were consistently administered in SC format for this group. In the illiterate group, both EQ-HWB and EQ-5D-5L were administered in IA format because of limited reading ability; WEMWBS was not administered to this group because of concerns about conceptual and linguistic

comprehension. In the patient group, the EQ-HWB was administered in both SC and IA formats, whereas the EQ-5D-5L and WEMWBS were administered in SC format.

Although the EQ-HWB and EQ-5D-5L instruments were designed to accommodate IA delivery, the WEMWBS do not have official IA versions. Nevertheless, for consistency across instruments and respondent groups, trained interviewers followed standardized procedures during IA sessions. They read all items, instructions, and response options exactly as written, avoiding interpretation or elaboration unless respondents explicitly requested clarification. This approach was used to maintain uniformity in administration and minimize interviewer influence.

In a small number of cases, primarily within the illiterate and patient groups, respondents asked for clarification on certain well-being-related items, particularly those concerning abstract or relational concepts such as "feeling supported" or "having control over daily life." When this occurred, interviewers were instructed to repeat the item verbatim and, if necessary, use a predefined example from the translation guide, without deviating from the original item wording. These instances were noted but were infrequent and did not lead to any protocol deviations or item exclusions.

In addition, all respondents completed a GH item to monitor perceived changes in health status between sessions. The GH question was administered once for each group, with the literate and illiterate groups completed it 2 weeks after baseline, whereas the patient group completed it 1 month after baseline, aligned with their hospital visit schedule. Responses to this item were used to identify participants with stable health status ("no change") for inclusion in test-retest reliability analyses.

Instruments

The questionnaire included standardized measures in the official Indonesian versions, presented in a fixed order, namely, EQ-HWB, EQ-5D-5L, EQ-VAS, and WEMWBS. Respondents were also asked to provide socio-demographic information (age, gender, education, employment, and disease). Study assistants administered the questionnaire (both IA and SC versions) and provided assistance when respondents faced any completion issues. In the follow-up questionnaire, respondents rated any changes in health status using the GH questionnaire.

EQ-HWB

EQ-HWB is a standardized tool designed to measure various aspects of health and well-being, featuring 25 items. A short version, EQ-HWB-9 (EQ Health and Well-being Short version), contains 9 items. This instrument evaluates a broad range of impacts, including those on the health and well-being of both care recipients and caregivers.¹ Items are assessed using 5-level response scales describing frequency, severity, or difficulty, with a recall period of "in the last 7 days." In addition, EQ-HWB-9, a subset of 9 items from EQ-HWB, primarily serves as a self-classifier for health valuations.

The translation process for EQ-HWB followed EQ-HWB Translation Guidelines provided by the EuroQol Version Management Committee. This rigorous process involved 4 steps: (1) forward translation and production of the first consensus version, (2) back translation and production of the second consensus version, (3) cognitive debriefing and production of the third consensus version, and (4) production and proofreading of the final Indonesian version. In this study, the 25-item EQ-HWB was administered, and responses were used to derive EQ-HWB-9. Level summary scores (LSS) were calculated for both EQ-HWB

and EQ-HWB-9 by summing the responses from the 25 and 9 items, respectively, with higher scores confirming worse health and well-being. The theoretical LSS ranges of 25-125 (EQ-HWB) and 9-45 (EQ-HWB-9) were rescaled to a 0 to 100 range to ensure comparability with other measures in the study. In addition, an index value for EQ-HWB-9 was derived using the UK pilot value set, with higher scores reflecting better health and well-being.¹² Similar to other EuroQol instruments, EQ-HWB-9 is expected to include preference-weighted value sets to support the economic evaluation of health and social care.¹³

EQ-5D-5L

The EQ-5D-5L is a measure of health-related quality of life consisting of 2 sections.¹⁴ The first section is a descriptive system with 5 dimensions, namely, mobility, self-care, usual activities, pain/discomfort, and anxiety/depression. Each dimension has 5 response levels, namely, no problems, slight problems, moderate problems, severe problems, and extreme problems/unable to. These responses combine into a 5-digit health state profile, for example, "11122" represents no problems in mobility, self-care, and usual activities, with slight problems in pain/discomfort and anxiety/depression. The descriptive system is evaluated by calculating a LSS, which sums the 5 digits of the health state profile. Higher scores show worse health-related quality of life. The theoretical LSS range of 5 to 25, is rescaled to a 0 to 100 range for analysis. In addition, each health state profile is assigned an index value using the Indonesian value set.¹⁵ The second section of the EQ-5D-5L is the visual analog scale (VAS), in which respondents rate general health status on a vertical scale from 0 ("worst imaginable health") to 100 ("best imaginable health"). The recall period for both sections of the EQ-5D-5L is "today."

WEMWBS

The WEMWBS is a measure of positive mental well-being consisting of 14 items. These items assess feelings such as optimism about the future, a sense of usefulness, relaxation, interest in others, self-esteem, closeness to others, being loved, interest in new things, confidence, cheerfulness, problem-solving ability, clear thinking, decisiveness, and having energy. WEMWBS uses a recall period of "over the last 2 weeks," with each item is rated on a 5-point scale, namely, none of the time, rarely, some of the time, often, and all of the time.¹⁶ The total WEMWBS score is calculated by summing the responses across all items, with a possible range from 14 to 70. Higher scores indicate greater mental well-being. In this study, the total WEMWBS score was used to assess convergent validity with EQ-HWB and EQ-HWB-9 scores in the literate and patient groups.

GH

To assess perceived changes in health status, respondents were asked a single-item question: "Overall, has there been any change in your health compared to the first time you saw us?" Respondents selected from 7 response options: much worse, moderately worse, slightly worse, no change, slightly better, moderately better, and much better. Responses were categorized into 3 groups: the first 3 options indicated a clinically meaningful deterioration, "no change" represented stable health, and the final 3 options were interpreted as a clinically meaningful improvement.¹⁷

Statistical analysis

Sample characteristics were summarized using descriptive statistics. Continuous variables were reported as means with standard deviations, whereas categorical variables were presented as counts with percentages. Moreover, frequencies,

interquartile ranges, means, and standard deviations for all domain responses across the measures were reported. The measurement properties of the outcome measures were subsequently assessed using ceiling and floor effects, test-retest reliability, convergent validity with corresponding domains of other measures, and known-group validity. All analyses were conducted using Stata 17.0,¹⁸ with results considered statistically significant at $P < .05$.

Ceiling and floor effects

Ceiling and floor effects were determined by assessing the proportion of respondents achieving the best and worst responses on each item and overall scores of the measures. A ceiling or floor effect was considered present when more than 70% or less than 5% of the respondents selected the top or bottom response option, respectively.³ In addition, the relative frequency of the most common health state profiles for EQ-HWB-9 and EQ-5D-5L was reported when described by at least 3 respondents.

Test-retest reliability

Test-retest reliability was analyzed for both the IA and SC versions. For the SC version, the analysis was conducted with literate and patient samples, whereas the IA version included all respondents (literate, illiterate, and patients). To identify respondents with stable health status across sessions, a GH transition item was used. This item asked participants to rate their overall health compared with the previous visit, and only those who reported "no change" were included in the test-retest reliability analysis. Reliability of EQ-HWB items was evaluated using Gwet's AC2,¹⁹ in which values interpreted as poor (<0.0), slight (0.0-0.19), fair (0.20-0.39), moderate (0.40-0.59), strong (0.60-0.79), and almost-perfect agreement (≥ 0.80).²⁰ Furthermore, intraclass correlation coefficients (ICC) were computed for instrument-level analysis, including EQ-HWB, EQ-HWB-9, and utility scores for the short version, using a 2-way mixed effects model with absolute agreement. ICC values were also interpreted as poor (0.0-0.39), fair (0.40-0.59), good (0.60-0.74), and excellent (0.75-1.0).²¹

Convergent validity

Convergent validity assessed the strength of relationships between measures of related constructs.²² Spearman's rank correlations were used for individual items or dimensions, whereas Pearson's correlation was applied for instrument-level analyses, including index values and LSS. Correlation coefficients were interpreted as weak ($r = 0.1-0.29$), moderate ($r = 0.30-0.49$), or strong ($r = 0.50$ and above).²³ It was hypothesized that strong correlations would exist between EQ-HWB-9 LSS, EQ-5D-5L LSS, EQ-VAS, and WEMWBS total score, between EQ-HWB LSS, EQ-5D-5L LSS, EQ-VAS, and WEMWBS total score, and between the Utility Score short version and WEMWBS total score. Moreover, at the item level, moderate to strong correlations were predicted between specific domains of EQ-HWB and corresponding items in the EQ-5D-5L.

Known-group validity

Known-group validity examined the ability of EQ-HWB LSS, EQ-HWB-9 LSS, and EQ-HWB-9 index value to distinguish between groups. Analyses compared patients vs nonpatients and respondents with EQ VAS score <80 and ≥ 80 using the Mann-Whitney U Test. It was hypothesized that patients and those with EQ-VAS scores <80 would report lower index scores on EQ-HWB-9, and higher LSS on EQ-HWB and EQ-HWB-9 compared with their counterparts.

Results

Characteristics of the Respondents

The final sample comprised 300 respondents. The mean age of the final respondents was 39.22 ± 15.81 , with 62.67% being female, 50.00% having attained secondary education, and 52.33% actively used (Table 1). In the illiterate group, most respondents had attained no formal education, primary education, or below. Patient group included equal proportions of individuals with diabetes and tuberculosis (50.00% each).

Summary Statistics

Regarding EQ-HWB LSS scores, the illiterate group had the lowest mean scores for both the short (12.50 ± 3.80) and long (34.84 ± 9.95) versions (Table 1). Conversely, patient group had the highest mean scores for both versions (EQ-HWB-9: 23.92 ± 6.23 ; EQ-HWB: 60.74 ± 17.01). For the short-version utility score, the illiterate group had the highest mean score (0.92 ± 0.08), whereas the patient group had the lowest (0.59 ± 0.24).

Ceiling and Floor

Regarding EQ-HWB-9 items, the proportion of respondents reporting the best responses ranged from 24.0% ("Exhaustion") to 71.67% ("Getting around inside and outside"), which specifically showed a clear ceiling effect. Respondents who reported the worst responses ranged from 0.67% (in items such as "Getting around inside and outside" and "Cognition") to 5.33% ("Exhaustion") (see Appendix Table 1 in Supplemental Materials).

Moreover, no EQ-HWB-9 items showed a clear floor effect (defined as $<5\%$). At the instrument level, 24 respondents (8.00%) reported the worst possible responses across all EQ-HWB-9 items.

For EQ-HWB items, respondents who reported the best responses ranged from 24.0% ("Exhaustion") to 86.33% ("Hearing"), with 5 out of 25 items showing a clear ceiling effect, namely "Hearing" (86.33%), "Personal care" (81.33%), "Frustration" (73.0%), "Getting around inside and outside" (71.67%), and "Nothing to look forward" (70.67%). The proportion reporting the worst responses ranged from 0.0% (eg, "Sight," "Hearing," "Personal care," "Memory," "Pain severity," and "Discomfort severity") to 6.33% ("Feel good about self" and "Do things one wanted to do"). No floor effect was observed for any EQ-HWB item. At the instrument level, 13 respondents (4.33%) reported the best possible responses across all EQ-HWB items.

Test-Retest Reliability

Test-retest reliability analyses were based on 123 respondents who reported no change in their general health status between sessions: 88 from the literate group, 25 from the illiterate group, and 10 from the patient group. At the item level, the EQ-HWB demonstrated good to excellent test-retest reliability across SC and IA modes in the literate, illiterate, and patient groups (Tables 2 and 3). Items related to physical and functional aspects, such as "Getting around inside and outside," "Personal care," "Hearing," and "Sight" consistently showed almost-perfect agreement ($AC2 \geq 0.85$) across all formats and populations. Interestingly, several psychosocial items also exhibited high

Table 1. Descriptive statistics.

Characteristics	Full sample (n = 300)		Literate (n = 200)		Illiterate (n = 50)		Patients (n = 50)	
	n	%	n	%	n	%	n	%
Age	M = 39.22, SD = 15.81		M = 34.00, SD = 13.98		M = 50.94, SD = 11.01		M = 48.40, SD = 16.38	
Gender								
Male	112	37.33	83	41.50	4	8.00	25	50.00
Female	188	62.67	117	58.50	44	88.00	25	50.00
Education								
Low	49	16.33	6	3.00	28	56.00	15	30.00
Middle	150	50.00	100	50.00	22	44.00	28	56.00
High	101	33.67	94	47.00	0	0.00	7	14.00
Employment Status								
Active	157	52.33	117	58.50	8	16.00	32	64.00
Inactive	105	35.00	46	23.00	42	84.00	17	34.00
Students	38	12.67	37	18.50	0	0.00	1	2.00
Disease								
Diabetes	25	8.33	0	0.00	0	0.00	25	50.00
TB	25	8.33	0	0.00	0	0.00	25	50.00
EQ-HWB/EQ-HWB-9	M	SD	M	SD	M	SD	M	SD
LSS EQ-HWB	44.47	15.98	42.81	13.99	34.84	9.95	60.74	17.01
LSS EQ-HWB-9	16.71	6.54	15.96	5.75	12.50	3.80	23.92	6.23
Utility Score EQ-HWB-9	0.82	0.20	0.85	0.16	0.92	0.08	0.59	0.24

Note. Education for the low category refers to no formal education or primary school level, middle refers to secondary school level, high refers to senior secondary level or above.

M indicates mean.

temporal stability. In particular, “Loneliness (S),” “Unsupported,” and “Frustration” demonstrated excellent ICC values in both IA (0.975, 0.973, and 0.981, respectively) and SC (0.921, 0.934, and 0.930, respectively) formats. This finding indicates that these items, despite involving subjective or emotional constructs, yielded highly consistent responses over time.

At the instrument level, ICCs further confirmed the reliability of EQ-HWB and EQ-HWB-9 scores over time (Table 3). The EQ-HWB LSS showed excellent reliability across all respondents in both SC (ICC = 0.975) and IA (ICC = 0.937)

formats. Similarly, the EQ-HWB-9 utility score yielded high in SC (ICC = 0.934) and IA (ICC = 0.934), indicating strong stability over time. However, subgroup analyses revealed slightly lower agreement levels in the illiterate group, although values remained within the acceptable good range. Specifically, the ICC for the EQ-HWB LSS was 0.755, for the EQ-HWB-9 LSS was 0.669, and for the EQ-HWB-9 utility score was 0.691. These findings suggest that although the instruments are reliable for use across literacy levels, response variability may be somewhat greater in low-literacy populations.

Table 2. The difference between EQ-HWB/EQ-HWB-9 via self-completion and interviewer administered.

EQ-HWB/EQ-HWB-9	All respondents (literate + patients) (n = 250)	Literate (n = 200)	Patients (n = 50)
	ICC (95% CI)		
LSS EQ-HWB	0.914 (0.891-0.932)	0.853 (0.810-0.887)	0.989 (0.980-0.994)
LSS EQ-HWB-9	0.922 (0.901-0.939)	0.868 (0.829-0.899)	0.988 (0.979-0.993)
Utility Score EQ-HWB-9	0.930 (0.910-0.945)	0.870 (0.832-0.900)	0.949 (0.840-0.978)
	Gwet's AC (95% CI)		
1. Sight	0.854 (0.804-0.904)	0.822 (0.762-0.883)	1.000 (1.000-1.000)
2. Hearing	0.854 (0.804-0.904)	0.938 (0.903-0.973)	1.000 (1.000-1.000)
3. Getting around inside and outside (EQ-HWB-9)	0.949 (0.920-0.977)	0.931 (0.893-0.968)	0.948 (0.874-1.000)
4. Day-to-day activities (EQ-HWB-9)	0.932 (0.898-0.966)	0.785 (0.722-0.848)	0.952 (0.884-1.000)
5. Personal care	0.921 (0.886-0.957)	0.902 (0.858-0.947)	1.000 (1.000-1.000)
6. Sleep	0.658 (0.589-0.727)	0.594 (0.512-0.675)	0.926 (0.841-1.000)
7. Exhaustion (EQ-HWB-9)	0.521 (0.446-0.595)	0.414 (0.329-0.500)	0.951 (0.883-1.000)
8. Loneliness (EQ-HWB-9)	0.733 (0.670-0.795)	0.691 (0.617-0.765)	0.904 (0.811-0.998)
9. Unsupported	0.709 (0.645-0.773)	0.661 (0.585-0.737)	0.907 (0.817-0.998)
10. Memory	0.605 (0.531-0.679)	0.547 (0.462-0.632)	0.846 (0.725-0.967)
11. Cognition (EQ-HWB-9)	0.530 (0.457-0.604)	0.443 (0.360-0.527)	0.871 (0.760-0.983)
12. Anxiety (EQ-HWB-9)	0.498 (0.423-0.573)	0.404 (0.319-0.488)	0.902 (0.806-0.998)
13. Unsafe	0.678 (0.611-0.745)	0.629 (0.551-0.708)	0.926 (0.841-1.000)
14. Frustration	0.617 (0.546-0.687)	0.569 (0.487-0.651)	0.830 (0.709-0.951)
15. Sadness or depression (EQ-HWB-9)	0.557 (0.484-0.630)	0.485 (0.402-0.569)	0.877 (0.772-0.983)
16. Nothing to look forward	0.748 (0.687-0.809)	0.724 (0.653-0.794)	0.856 (0.744-0.969)
17. No control over daily life (EQ-HWB-9)	0.668 (0.601-0.735)	0.616 (0.537-0.694)	0.902 (0.806-0.997)
18. Coping	0.781 (0.723-0.839)	0.748 (0.681-0.816)	0.928 (0.845-1.000)
19. Accepted by others (R)	0.538 (0.464-0.612)	0.445 (0.360-0.530)	0.928 (0.845-1.000)
20. Feel good about self (R)	0.592 (0.519-0.664)	0.525 (0.441-0.609)	0.877 (0.772-0.983)
21. Do things one wanted to do (R)	0.546 (0.472-0.621)	0.438 (0.353-0.524)	1.000 (1.000-1.000)
22. Pain (frequency)	0.568 (0.495-0.642)	0.481 (0.396-0.566)	0.947 (0.873-1.000)
23. Pain (severity) (EQ-HWB-9)	0.698 (0.634-0.763)	0.621 (0.540-0.701)	0.949 (0.878-1.000)
24. Discomfort (frequency)	0.650 (0.582-0.719)	0.578 (0.497-0.659)	0.951 (0.882-1.000)
25. Discomfort (severity)	0.707 (0.641-0.774)	0.653 (0.574-0.731)	0.948 (0.874-1.000)

ICC indicates intraclass correlation coefficient; LSS, level summary score; R, reversed item.

Table 3. Head-to-head comparison of EQ-HWB/EQ-HWB-9 test-retest via self-completion (SC) vs interviewer administered (IA).

EQ-HWB/EQ-HWB-9	All respondents (n = 123)		Literate (n = 88)		Illiterate (n = 25)	Patients (n = 10)	
	SC (literate + patients)	IA (literate + illiterate + patients)	SC	IA	IA	SC	IA
ICC							
LSS EQ-HWB	0.934	0.934	0.911	0.911	0.669	0.988	0.988
LSS EQ-HWB-9	0.975	0.937	0.916	0.914	0.755	0.987	0.990
Utility Score EQ-HWB-9	0.934	0.934	0.911	0.911	0.669	0.988	0.988
Gwet's AC							
1. Sight	0.975	0.971	0.973	0.962	0.976	1.000	1.000
2. Hearing	0.985	0.995	0.984	0.993	1.000	1.000	1.000
3. Getting around inside and outside (EQ-HWB-9)	0.995	0.995	0.996	0.996	0.834	0.940	0.940
4. Day-to-day activities (EQ-HWB-9)	0.981	0.982	0.982	0.979	0.906	0.981	0.981
5. Personal care	0.993	0.995	0.992	0.992	1.000	1.000	1.000
6. Sleep	0.939	0.927	0.942	0.909	0.902	0.931	0.931
7. Exhaustion (EQ-HWB-9)	0.844	0.842	0.853	0.839	0.907	0.832	0.832
8. Loneliness (EQ-HWB-9)	0.921	0.975	0.922	0.967	0.958	0.953	0.977
9. Unsupported	0.934	0.973	0.928	0.933	0.989	0.988	0.988
10. Memory	0.904	0.897	0.893	0.891	0.814	1.000	1.000
11. Cognition (EQ-HWB-9)	0.931	0.927	0.928	0.922	0.826	0.970	0.970
12. Anxiety (EQ-HWB-9)	0.856	0.930	0.854	0.936	0.803	1.000	1.000
13. Unsafe	0.913	0.945	0.920	0.933	0.946	0.913	0.957
14. Frustration	0.930	0.981	0.927	0.973	1.000	0.983	0.982
15. Sadness or depression (EQ-HWB-9)	0.899	0.959	0.898	0.950	0.951	0.979	1.000
16. Nothing to look forward	0.939	0.927	0.951	0.925	0.930	0.811	0.785
17. No control over daily life (EQ-HWB-9)	0.890	0.933	0.889	0.908	0.955	0.963	0.929
18. Coping	0.976	0.966	0.979	0.954	0.951	0.896	0.983
19. Accepted by others (R)	0.833	0.813	0.831	0.818	0.879	0.905	0.878
20. Feel good about self (R)	0.812	0.870	0.808	0.855	0.941	0.967	0.915
21. Do things one wanted to do (R)	0.866	0.823	0.863	0.833	0.891	0.925	0.925
22. Pain (frequency)	0.905	0.871	0.912	0.851	0.920	0.850	0.850
23. Pain (severity) (EQ-HWB-9)	0.934	0.934	0.937	0.944	0.814	0.932	0.855
24. Discomfort (frequency)	0.920	0.918	0.918	0.915	0.912	0.921	0.760
25. Discomfort (severity)	0.934	0.930	0.849	0.884	0.691	0.970	0.970

ICC indicates intraclass correlation coefficient; LSS, level summary score; R, reversed item.

Convergent Validity

Strong correlations were observed at the instrument level between measures of closely related constructs in the full sample, namely, EQ-HWB-9 LSS and EQ-5D-5L LSS ($r = 0.83$), EQ-HWB LSS and EQ-5D-5L LSS ($r = 0.83$), EQ-HWB-9 LSS and EQ-VAS ($r = -0.69$), EQ-HWB LSS and EQ-VAS ($r = -0.77$), EQ-HWB-9 LSS and WEMWBS ($r = -0.59$), EQ-HWB LSS and WEMWBS ($r = -0.59$), and EQ-HWB-9 utility score and WEMWBS ($r = 0.57$). Moderate to strong correlations were also found between EQ-5D-5L LSS and EQ-VAS across the literate,

illiterate, and patient groups. However, EQ-HWB-9 LSS and EQ-HWB LSS did not show significant correlations with EQ-VAS (Table 4).

As hypothesized, moderate to strong correlations were found between conceptually overlapping items in the full sample, namely, EQ-HWB getting around inside and outside and EQ-5D-5L mobility ($r = 0.77$), EQ-HWB anxiety and EQ-5D-5L anxiety/depression ($r = 0.70$), EQ-HWB sadness or depression and EQ-5D-5L anxiety/depression ($r = 0.69$), EQ-HWB personal care and EQ-5D-5L self-care ($r = 0.60$), EQ-HWB day-to-day activities and EQ-5D-5L usual

Table 4. Correlations between EQ-HWB, EQ-HWB-9, EQ-5D-5L, and WEMWBS.

	All respondents (n = 300)	Literate (n = 200)	Illiterate (n = 50)	Patients (n = 50)
Pearson's correlation	Correlation coefficient			
EQ-HWB-9 LSS ~ EQ-5D-5L LSS	0.827*	0.765*	0.468*	0.863*
EQ-HWB LSS ~ EQ-5D-5L LSS	0.828*	0.749*	0.526*	0.902*
EQ-HWB-9 LSS ~ EQ-VAS	−0.686*	−0.629*	−0.208	−0.589*
EQ-HWB LSS ~ EQ-VAS	−0.769*	−0.669*	−0.256	−0.632*
EQ-HWB-9 LSS ~ WEMWBS-ID	-	−0.585*	-	-
EQ-HWB LSS ~ WEMWBS-ID		−0.586*		
EQ-HWB-9 Utility Score ~ WEMWBS-ID		0.568*		
Spearman's correlation	Correlation coefficient			
EQ-HWB-9 getting around inside and outside ~ EQ-5D-5L mobility	0.766*	0.636*	0.192	0.876*
EQ-HWB-9 day-to-day activities ~ EQ-5D-5L usual activities	0.573*	0.496*	0.448*	0.669*
EQ-HWB personal care ~ EQ-5D-5L self-care	0.603*	0.610*	0.474*	0.699*
EQ-HWB-9 loneliness ~ EQ-5D-5L anxiety/depression	0.431*	0.433*	0.095	0.381*
EQ-HWB-9 anxiety ~ EQ-5D-5L anxiety/depression	0.704*	0.608*	0.671*	0.867*
EQ-HWB unsafe ~ EQ-5D-5L anxiety/depression	0.436*	0.291*	0.203	0.709*
EQ-HWB frustration ~ EQ-5D-5L anxiety/depression	0.519*	0.412*	0.367*	0.759*
EQ-HWB-9 sadness or depression ~ EQ-5D-5L anxiety/depression	0.689*	0.653*	0.501*	0.772*
EQ-HWB nothing to look forward ~ EQ-5D-5L anxiety/depression	0.508*	0.459*	0.148	0.703*
EQ-HWB pain (frequency) ~ EQ-5D-5L pain/discomfort	0.555*	0.508*	0.433*	0.630*
EQ-HWB-9 pain (severity) ~ EQ-5D-5L pain/discomfort	0.548*	0.529*	0.250	0.670*

* $P < .100$.

activities ($r = 0.57$), EQ-HWB pain (frequency) and EQ-5D-5L pain/discomfort ($r = 0.56$), EQ-HWB pain (severity) and EQ-5D-5L pain/discomfort ($r = 0.55$), EQ-HWB frustration and EQ-5D-5L anxiety/depression ($r = 0.52$), EQ-HWB nothing to look forward and EQ-5D-5L anxiety/depression ($r = 0.51$), EQ-HWB unsafe and EQ-5D-5L anxiety/depression ($r = 0.44$), and EQ-HWB loneliness and EQ-5D-5L anxiety/depression ($r = 0.43$) (Table 4).

Moderate to strong correlations were found in the literate and patient groups when analyzed by sample groups, with the exception of EQ-HWB unsafe and EQ-5D-5L anxiety/depression, which had a weak correlation in the literate group. Conversely, fewer conceptually overlapping items were correlated in the illiterate group. Unlike the literate and patient groups, no correlations were found between EQ-HWB getting around inside and outside and EQ-5D-5L mobility, EQ-HWB loneliness and EQ-5D-5L anxiety/depression, EQ-HWB unsafe and EQ-5D-5L anxiety/depression, EQ-HWB nothing to look forward to and EQ-5D-5L anxiety/depression, as well as EQ-HWB pain (severity) and EQ-5D-5L pain/discomfort in the illiterate group.

Known-Group Validity

EQ-HWB and EQ-HWB-9 showed statistically significant discrimination across groups based on EQ-VAS scores and patients vs nonpatients status (Table 5). When grouping individuals based on EQ-VAS scores in the full sample, those with EQ-VAS < 80 had significantly lower EQ-HWB-9 utility scores and higher LSS EQ-HWB and EQ-HWB-9 ($P < .001$). Similar results were observed when comparing patients to nonpatients ($P < .001$).

Discussion

To our knowledge, this is the first study to validate EQ-HWB and EQ-HWB-9 by comparing SC and IA methods across both the general public and patient groups, while accounting for literacy status. Furthermore, it provided evidence on the test-retest reliability and validity of these measures. The results showed strong psychometric performance of the Indonesian

Table 5. Comparison of EQ-HWB/EQ-HWB-9 scores based on known groups.

Based on EQ-VAS (<80 and ≥80)					
Mann-Whitney U test	U	z	P	n EQ-VAS < 80 = 86 n EQ-VAS ≥ 80 = 214	
LSS EQ-HWB	2753.00	−9.495	<.001		
LSS EQ-HWB-9	3108.00	−8.991	<.001		
Utility Score	2873.50	9.318	<.001		
EQ-HWB-9					
Based on patients status					
Mann-Whitney U test	U	z	P	n patients = 50 n nonpatients = 250	
LSS EQ-HWB	1982.00	−7.625	<.001		
LSS EQ-HWB-9	1567.50	−8.221	<.001		
Utility Score	1544.50	−8.407	<.001		
EQ-HWB-9					

versions of EQ-HWB and EQ-HWB-9 in terms of validity and reliability. Only 1 item in EQ-HWB-9 had a low ceiling effect, and none showed a floor effect. In contrast, EQ-HWB had more items with ceiling effects (5 out of 25). This difference could be attributed to the wider range of items in EQ-HWB, whereas EQ-HWB-9, with its narrower item range, was less prone to ceiling and floor effects.²⁴

Items such as “Hearing,” “Personal care,” “Frustration,” and “Nothing to look forward to” might show ceiling effects because respondents from the general public were relative healthy, making them less likely experience significant challenges in these areas more likely to report the most favorable responses. These patterns are consistent with earlier studies validating the EQ-HWB and EQ-HWB-9, which also reported ceiling effects in certain items when used in general population samples.^{25,26} For example, 1 Italian study observed that social or emotional well-being items, such as “Accepted by others,” tended to show skewed responses in healthier populations.²⁵ Similarly, ceiling effects in functional domains were noted in studies comparing EQ-HWB with EQ-5D-5L, particularly where the general population sample had limited health impairments.^{27,28} Therefore, these findings suggest that both versions are suitable for capturing changes in health and well-being status within the general population.

The test-retest analysis demonstrated excellent reliability for both EQ-HWB and EQ-HWB-9 across SC and IA formats. At the item level, strong test-retest reliability was observed for most dimensions, particularly those assessing physical and functional aspects (eg, “Getting around inside and outside,” “Hearing,” “Sight,” and “Personal care”), which consistently showed almost-perfect agreement across groups and modes. These domains reflect observable and relatively stable aspects of health, which are less subject to momentary mood or cognitive fluctuations.^{29,30} Comparable results have been reported in previous validation studies, including in Australia and the United Kingdom, where EQ-HWB and EQ-HWB-9 items related to mobility, sensory functions, and self-care also demonstrated high test-retest reliability, particularly among older adults and caregivers with chronic conditions.^{31,32} This consistency reinforces the capacity of EQ-HWB instruments to reliably capture physical and sensory impairments over time. It is especially valuable in longitudinal studies or evaluations of intervention outcomes, in which test-retest stability ensures that observed changes in scores reflect genuine health shifts rather than measurement error.³³ Moreover, the strong agreement observed in both SC and IA formats supports the flexible application of the EQ-HWB tools across populations with varying literacy levels. This aligns with finding from earlier studies suggesting that interviewer administration, when standardized, does not compromise psychometric robustness and can enhance accessibility for underserved groups.³⁴

Interestingly, several psychosocial items, including “Loneliness (S),” “Unsupported,” and “Frustration” exhibited excellent test-retest reliability, with ACs exceeding 0.92 across both IA and SC formats. Notably, other emotionally sensitive items, such as “Anxiety,” “Exhaustion,” and “Cognition” (also included in EQ-HWB-9) similarly demonstrated excellent agreement. These results indicate that, despite their subjective and potentially fluctuating nature, these well-being domains can yield highly consistent responses over time when administered with a 7-day recall period. This pattern contrasts with prior findings in which psychological constructs often showed lower stability due to inherent within-person variability, particularly with longer recall intervals.^{25,29} Our findings suggest that the EQ-HWB items may be well formulated and culturally comprehensible enough to

capture stable perceptions of psychosocial health, reinforcing their reliability for use in repeated-measures studies and longitudinal monitoring. Overall, the results provide strong evidence that both EQ-HWB and EQ-HWB-9 are reliable over time, with excellent performance even in SC vs IA comparisons and across literacy levels. The high ACs observed for psychosocial items reinforce the potential of these instruments to capture a broad and stable profile of well-being across diverse respondent groups.

At the instrument level, both EQ-HWB and EQ-HWB-9 showed excellent reliability, with ICCs exceeding 0.93 across formats. These findings are in line with prior validation studies, which reported favorable test-retest performance of EQ-HWB-9, albeit often in smaller or more homogeneous samples (eg, caregivers or patients with cancer).^{26,31,32,35} Previous study also reported that EQ-HWB-9 performed comparably with EQ-5D-5L and FACT-G/FACT-8D in oncology settings.³⁵ Our study expands upon this evidence by demonstrating high reliability in more heterogeneous groups, including respondents from the general population, patients, and notably, those with low literacy. Moreover, subgroup analyses revealed that although all ICC values in the illiterate group remained within the acceptable range, they were relatively lower compared with other groups (eg, EQ-HWB LSS = 0.755; EQ-HWB-9 LSS = 0.669; EQ-HWB-9 utility = 0.691). This suggests a modest increase in response variability among low-literacy populations, despite the use of IA methods. Nevertheless, the overall stability across formats and populations reinforces the robustness and adaptability of the EQ-HWB instruments for use in diverse health system and cultural contexts, including settings where literacy and mode of administration may vary.

Strong convergence between EQ-HWB/EQ-HWB-9, EQ-5D-5L, and WEMWBS supports the construct validity of EQ-HWB/EQ-HWB-9. This alignment was evident in several conceptually overlapping items across the measures. The strongest correlation was observed between the “getting around inside and outside” dimension of EQ-HWB and the mobility dimension of the EQ-5D-5L, followed by the correlation between EQ-HWB anxiety and EQ-5D-5L anxiety/depression. Specifically, the composite dimensions of the EQ-5D-5L had varying degrees of relationship with individual EQ-HWB dimensions, showing that EQ-HWB might offer better clarity in capturing respondents’ experiences.^{5,28,35} For example, although EQ-5D-5L anxiety/depression strongly correlated with EQ-HWB anxiety, it only moderately correlated with EQ-HWB loneliness and EQ-HWB unsafe. A similar pattern of divergence was also observed in the Italian EQ-HWB validation study conducted with a general population sample.⁵

Known-groups comparisons demonstrated that both the EQ-HWB and EQ-HWB-9 significantly differentiated between subgroups with expected differences in health status, supporting their known-groups validity. In particular, both instruments discriminated well between respondents reporting EQ-VAS scores below 80 and those with scores of 80 or above, indicating sensitivity to self-perceived health status. Furthermore, both versions successfully distinguished between patient and non-patient groups, reinforcing their ability to capture the more complex health and well-being dynamics experienced by individuals with chronic conditions. These differences may be attributed to the broader symptom burden, reduced functional capacity, and psychosocial challenges often faced by individuals living with chronic illnesses, such as fatigue, pain, or limited mobility, which are captured by the multidimensional content of EQ-HWB.²⁹ Additionally, those reporting higher EQ-VAS scores likely experienced fewer physical or emotional limitations, leading to systematically more favorable responses across EQ-

HWB domains. These findings are consistent with previous validation studies, in which the EQ-HWB and EQ-HWB-9 also showed satisfactory known-groups performance in differentiating between general population and clinical samples and among those with and without long-term conditions.^{35,36} This evidence highlights the robustness of EQ-HWB instruments in detecting known differences across populations, even when applied in diverse cultural and health system contexts, such as Indonesia.

Study Limitations

This study had some limitations. First, the IA survey administration might have influenced responses because of potential social desirability or acquiescence bias, which could be reflected in the general good health status reported by patients. It was also important to consider that Asian patients were often less likely to report health problems, including physical and mental symptoms, compared with their Western counterparts.^{37,38} Second, the measures in the survey were administered in a fixed order. This could potentially introduce bias, previous studies showed presentation order was unlikely to significantly affect responses or might have only a small effect when present.^{39,40} Third, because the analyses were based on self-reported respondent characteristics and medical conditions, reporting bias was likely to have occurred. Fourth, although the patient group included individuals with confirmed clinical diagnoses and a total sample size of 50 (exceeding the COSMIN-recommended minimum of 30 for known-groups testing),²² it comprised 2 different conditions, diabetes and tuberculosis, with 25 participants each. These conditions were chosen to reflect contrasting physical health burdens typically found in outpatient settings in Indonesia. However, the small sample sizes for each subgroup were not intended for direct comparisons between patient conditions and may limit the interpretability of subgroup-level analyses. Lastly, the sample size for the literate group was deliberately larger than other groups to serve as the main comparison group for both known-groups validity and mode of administration testing. This reflects our hypothesis that patients would show significantly worse outcomes than the general population and that literacy may influence response patterns. This imbalance in group sizes should be taken into account when interpreting between-group comparisons.

Conclusions

In conclusion, this study provided valuable evidence on the reliability and validity of both the SC and IA versions of the EQ-HWB and EQ-HWB-9 across literate, illiterate, and patient populations. The results demonstrated high agreement between SC and IA versions, supporting their interchangeability in many contexts. Notably, the IA administration of the EQ-HWB and EQ-HWB-9 was found to be feasible and psychometrically robust even in populations with low literacy levels. These findings endorse the broader applicability of the EQ-HWB instruments as outcome measures in diverse clinical and public health settings.

Author Disclosures

Author disclosure forms can be accessed below in the [Supplemental Material](#) section.

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REFERENCES

1. Brazier J, Peasgood T, Mukuria C, et al. The EQ-HWB: overview of the development of a measure of health and wellbeing and key results. *Value Health*. 2022;25(4):482–491.
2. Peasgood T, Mukuria C, Carlton J, et al. What is the best approach to adopt for identifying the domains for a new measure of health, social care and carer-related quality of life to measure quality-adjusted life years? Application to the development of the EQ-HWB? *Eur J Health Econ*. 2021;22(7):1067–1081.
3. Peasgood T, Mukuria C, Brazier J, et al. Developing a new generic health and wellbeing measure: psychometric survey results for the EQ-HWB. *Value Health*. 2022;25(4):525–533.
4. Augustovski F, Argento F, Rodríguez RB, Gibbons L, Mukuria C, Belizán M. The development of a new international generic measure (EQ-HWB): face validity and psychometric stages in Argentina. *Value Health*. 2022;25(4):544–557.
5. Masutti S, Falivena C, Purba FD, Jommi C, Mukuria C, Finch AP. Content validity of the EQ-HWB and EQ-HWB-9 in a sample of Italian patients, informal caregivers and members of the general public. *J Patient Rep Outcomes*. 2024;8(1):36.
6. Carlton J, Peasgood T, Mukuria C, et al. Generation, selection, and face validation of items for a new generic measure of quality of life: the EQ-HWB. *Value Health*. 2022;25(4):512–524.
7. Long C, Mao Z, Yang Z. A head-to-head comparison of EQ health and wellbeing and EQ-5D-5L in patients, carers, and general public in China. *Value Health*. 2024;27(7):848–856.
8. Bicar BB, Rubio MT, Barroso DA. Why is literacy not always a guarantee for quality of life. *J Higher Educ*. 2017;2(2):48–57.
9. Regmi KD. Global construction of literacy policies for “least developed countries”: focus on Ethiopia, Nepal, and Sierra Leone. *Adult Educ Q*. 2019;69(3):225–246.
10. UNESCO UIS. Indonesia. United Nations Educational, Scientific and Cultural Organization (UNESCO). <http://uis.unesco.org/en/country/id#slideoutsearch>. Accessed August 30, 2024.
11. Khozanatua F, Setiyani R, Kusumawardani LH. Predictors of COVID-19 related health literacy among older people living in rural areas of Indonesia. *Invest Educ Enferm*. 2023;41(2):e13.
12. Mukuria C, Peasgood T, McDool E, Norman R, Rowen D, Brazier J. Valuing the EQ health and wellbeing short using time trade-off and a discrete choice experiment: a feasibility study. *Value Health*. 2023;26(7):1073–1084.
13. EuroQol. EQ-HWB. <https://euroqol.org/research-at-euroqol/eq-hwb/>. Accessed August 30, 2024.
14. Herdman M, Gudex C, Lloyd A, et al. Development and preliminary testing of the new five-level version of EQ-5D (EQ-5D-5L). *Qual Life Res*. 2011;20(10):1727–1736.

15. Purba FD, Hunfeld JAM, Iskandarsyah A, et al. The Indonesian EQ-5D-5L value set. *Pharmacoeconomics*. 2017;35(11):1153–1165.
16. Tennant R, Hiller L, Fishwick R, et al. The Warwick-Edinburgh mental well-being scale (WEMWBS): development and UK validation. *Health Qual Life Outcomes*. 2007;5:1–13.
17. Uwer L, Rotonda C, Guillemin F, et al. Responsiveness of EORTC QLQ-C30, QLQ-CR38 and FACT-C quality of life questionnaires in patients with colorectal cancer. *Health Qual Life Outcomes*. 2011;9:1–10.
18. Stata 17. StataCorp LLC. <https://www.stata.com/stata-news/news36-2/>. Accessed August 30, 2024.
19. Gwet KL. Handbook of Inter-rater Reliability: The Definitive Guide to Measuring the Extent of Agreement Among Raters. Houston, TX: Advanced Analytics. LLC; 2014.
20. Landis JR, Koch GG. The measurement of observer agreement for categorical data. *Biometrics*. 1977;33(1):159–174.
21. Cicchetti DV. Interrater reliability standards in psychological evaluations. *Psychol Assess*. 1994;6(4):284–290.
22. Terwee CB, Bot SDM, de Boer MR, et al. Quality criteria were proposed for measurement properties of health status questionnaires. *J Clin Epidemiol*. 2007;60(1):34–42.
23. Cohen J. *Statistical Power Analysis for the Behavioral Sciences*. 11. Abingdon, United Kingdom: Routledge; 2013.
24. Murugappan MN, King-Kallimanis BL, Mangir C, et al. Floor and ceiling effects in the EORTC QLQ-C30 Physical Functioning Subscale among patients with advanced or metastatic breast cancer. *Cancer*. 2022;128(4):808–818.
25. Kuharić M, Pickard AS, Mukuria C, Finch AP. The measurement properties of the EQ-HWB and the EQ-HWB-9 Italian Population: A Comparative Study With EQ-5D-5L. *Value Health*. 2024;27(7):955–966.
26. Kuharic M, Mulhern B, Sharp LK, Turpin RS, Pickard AS. Comparison of EQ health and well-being long and short with other preference-based measures among United States informal caregivers. *Value Health*. 2024;27(7):967–977.
27. Kuharic M, Monteiro A, Pickard A. PCR255 the EQ health and wellbeing (EQ-HWB) and the EQ health and wellbeing short (EQ-HWB-9) as an outcome measure in mental health and well-being: A comparison to the EQ-5D-5L, EQ-5D-3L and the Short Warwick-Edinburgh mental well-being scale. *Value Health*. 2022;25(12):S439.
28. Monteiro AL, Kuharic M, Pickard AS. A comparison of a preliminary version of the EQ-HWB short and the 5-level version EQ-5D. *Value Health*. 2022;25(4):534–543.
29. Xu RH, Rencz F. Psychometric Properties of the Modified EQ Health and Wellbeing and its Shorter Version in a General Chinese Population: A Comparative Analysis With EQ-5D-5L, SF-6Dv2, ICEpop CAPability measure for Adults, Short Version of Warwick-Edinburgh Mental Wellbeing Scale, and World Health Organization Well-Being Index 5-Item. *Value Health*. 2025;28(8):1241–1252.
30. Feng YS, Kohlmann T, Janssen MF, Buchholz I. Psychometric properties of the EQ-5D-5L: a systematic review of the literature. *Qual Life Res*. 2021;30(3):647–673.
31. Bailey C, Peasgood T, Michalowsky B, Engel L. The psychometric performance of the EQ health and wellbeing in caregivers of persons living with dementia. *Value Health*. 2025;28(8):1221–1230.
32. Lee P, Engel L, Lubetkin E, Gao L. Exploring the comparability between EQ-5D and the EQ health and wellbeing in the general Australian population. *Value Health*. 2024;27(4):508–517.
33. Sousa H, Ribeiro O, Aleixo B, Christensen AJ, Figueiredo D. Evaluating the psychometric properties of patient-reported outcome measures for assessing symptoms in hemodialysis: A systematic review using COSMIN guidelines. *J Pain Symptom Manag*. 2024;67(1):e34–e57.
34. Finch AP, Meregaglia M, Ciani O, Roudijk B, Jommi C. An EQ-5D-5L value set for Italy using videoconferencing interviews and feasibility of a new mode of administration. *Soc Sci Med*. 2022;292:114519.
35. Pangestu S, Purba FD, Setyowibowo H, Azhar Y, Mukuria C, Rencz F. The psychometric properties of the EQ-HWB and EQ-HWB-9 patients with breast cancer: a comparative analysis with EQ-5D-5L, FACT-8D, and SWEMWBS. *Value Health*. 2025;28(3):449–459.
36. Kinchin I, Engel L, Rencz F. A comparative study of health and well-being measures in Ireland using EQ Health and Wellbeing (EQ-HWB) and its short version, EQ-5D-5L, and ICEpop Capability Measure for Adults (ICECAP-A). *Value Health*. 2025;28(8):1268–1279.
37. Birtel MD, Mitchell BL. Cross-cultural differences in depression between White British and South Asians: causal attributions, stigma by association, discriminatory potential. *Psychol Psychother Theor Res Pract*. 2023;96(1):101–116.
38. Chen CH, Tang ST, Chen CH. Meta-analysis of cultural differences in Western and Asian patient-perceived barriers to managing cancer pain. *Palliat Med*. 2012;26(3):206–221.
39. Cheung YB, Wong LC, Tay MH, et al. Order effects in the assessment of quality of life in cancer patients. *Qual Life Res*. 2004;13(7):1217–1223.
40. Childs AL, Submacular Surgery Trials Research Group. Effect of order of administration of health-related quality of life interview instruments on responses. *Qual Life Res*. 2005;14(2):493–500.